Practical No.1

Q. A) Write a python program for creating a series from scalar value.

Program:

```
import pandas as pd
s = pd.Series(50, index =[0, 1, 2, 3, 4])
print(s)
```

Output:

```
0    50
1    50
2    50
3    50
4    50
dtype: int64
```

B) Write a python program for creating a series from a dictionary.

Program:

```
import pandas as pd
d = {'Name':'Hardik', 'IplTeam':'MI', 'Runs': 1500}
s = pd.Series(d)
print(s)
```

Output:

```
Name        Hardik
IplTeam         MI
Runs          1500
dtype: object
```

Practical No.2

Q. A) Write a python program for Mathematical operations in Series.

Program:

```
import pandas as pd
s = pd.Series([1, 2, 3, 4, 5])
print('To Multiply all values in a series by 2')
print('---------------------------------------')
print(s*2)
print('To find the square of all values in a series')
print('---------------------------------------')
print(s**2)
print('To print all the values in a series that are greater than 2')
print('---------------------------------------')
print(s[s>2])
```

Output:

```
To Multiply all values in a series by 2
---------------------------------------
0     2
1     4
2     6
3     8
4    10
dtype: int64
To find the square of all values in a series
---------------------------------------
0     1
1     4
2     9
3    16
4    25
dtype: int64
To print all the values in a series that are greater than 2
---------------------------------------
2    3
3    4
4    5
dtype: int64
```

B) Write a python program for Mathematical operations adding two or more Series.

Program:

```
import pandas as pd
s1=pd.Series([1,2,3,4,5], index=['a','b','c','d','e'])
s2=pd.Series([10,20,30,40,50], index=['a','b','c','d','e'])
s3=pd.Series([5,14,23,32], index=['a','b','c','d'])
print('To Add series1 & series2')
print('--------------------------------')
print(s1+s2)
print('To Add series2 & series3')
print('--------------------------------')
print(s2+s3)
print('To Add series2 & series3 and Filled non matching index with 0')
print('--------------------------------')
print(s2.add(s3, fill_value=0))
```

Output:

```
To Add series1 & series2
--------------------------------
a    11
b    22
c    33
d    44
e    55
dtype: int64
To Add series2 & series3
--------------------------------
a    15.0
b    34.0
c    53.0
d    72.0
e     NaN
dtype: float64
To Add series2 & series3 and Filled non matching index with 0
--------------------------------
a    15.0
b    34.0
c    53.0
d    72.0
e    50.0
dtype: float64
```

Practical No.3

Q. Write a python program for Indexing in Series.

Program:

```
import pandas as pd
import numpy as np
arr=np.array(['a', 'b', 'c', 'd'],)
s=pd.Series(arr, index=['first', 'second', 'third', 'fourth'])
print(s)
# To print only indexes in series
print('\n indexes in Series are:::')
print(s.index)
```

Output:

```
first     a
second    b
third     c
fourth    d
dtype: object

 indexes in Series are:::
Index(['first', 'second', 'third', 'fourth'], dtype='object')
```

Date: 19/08/2026

Practical No.4

Q. Write a python program for Slicing in Series.

Program:

```
import pandas as pd
import numpy as np
arr=np.array([10,15,18,22,55,77])
s = pd.Series(arr, index=['A', 'B', 'C', 'D', 'E', 'F'])
print(s)
print(s[1:5:2])
print(s[0:6:2])
```

Output:

```
A    10
B    15
C    18
D    22
E    55
F    77
dtype: int32
B    15
D    22
dtype: int32
A    10
C    18
E    55
dtype: int32
```

Date: 10/09/2026

Practical No.5

Q. Write a python program for Slicing in dictionary.

Program:

```
import pandas as pd
dic1= { 'id': ['1', '2', '3', '4', '5'], 'Value1': ['A', 'C', 'E', 'G', 'I'],
'Value2': ['B', 'D', 'F', 'H', 'J']}
dic2= {'id': ['2', '3', '6', '7', '8'], 'Value1': ['K', 'M', '0', 'Q', 'S'],
'Value2': ['L', 'N', 'P', 'R', 'T']}
df1=pd.DataFrame(dic1)
df2=pd.DataFrame(dic2)
df3=pd.merge(df1, df2, left_on='id', right_on='id',
             how='outer', suffixes=('_left','_right'))
print(df3)
```

Output:

	id	Value1_left	Value2_left	Value1_right	Value2_right
0	1	A	B	NaN	NaN
1	2	C	D	K	L
2	3	E	F	M	N
3	4	G	H	NaN	NaN
4	5	I	J	NaN	NaN
5	6	NaN	NaN	0	P
6	7	NaN	NaN	Q	R
7	8	NaN	NaN	S	T

Practical No.6

Q. Write a Python program to perform read and write operations on a file.

Program:

A) Open a file and write data from the user and read data.

```
str =input("Enter the data into a file: ")
f1=open('abc.txt', 'w')
f1.write(str)
f1.close()
f2=open('abc.txt','r')
data=f2.read()
print(data)
f2.close()
```

Output:

```
Enter the data into a file: Python Programming Lab Mannual
Python Programming Lab Mannual
```

B) Open a file and write data into the file and read data.

```
filename = input("Enter the file name: ")
f1=open(filename, 'w')
f1.write('Computer science Engineering\n')
f1.write('Elecronics and Communication Engineering\n')
f1.write('Civil Engineering\n')
f1.close()
f2=open(filename, 'r')
data=f2.read()
print(data)
f2.close()
```

Output:

```
Enter the file name: Engineering Fields
Computer science Engineering
Elecronics and Communication Engineering
Civil Engineering
```

C) Open a file and read a particular line in the file.

```
filename = input("Enter the file name: ")
f1=open(filename, 'w')
f1.write('Computer science Engineering\n')
f1.write('Elecronics and Communication Engineering\n')
f1.write('Civil Engineering\n')
f1.close()
f2=open(filename, 'r')
data=f2.readline()
print(data)
f2.close()
```

Output:

```
Enter the file name: Engineering Fields
Computer science Engineering
```

Practical No.7

Q. Write a Python program to copy the contents of one file to another file.

Program:

```
filename1 = input("Enter the file name 1: ")
f1 = open(filename1, 'w')
f1.write('Computer Science Engineering\n')
f1.write('Electronics and Communication Engineering\n')
f1.write('Civil Engineering\n')
f1.close()
f2 = open(filename1, 'r')
data1 = f2.read()
print(data1)
f2.close()
filename2 = input("Enter the file name 2: ")
f3 = open(filename2, 'w')
f3.write(data1)
f3.close()
f3 = open(filename2, 'r')
data2 = f3.read()
print(data2)
f3.close()
```

Output:

```
Enter the file name 1: MCA
Computer Science Engineering
Electronics and Communication Engineering
Civil Engineering

Enter the file name 2: Engineering
Computer Science Engineering
Electronics and Communication Engineering
Civil Engineering
```

Date: 15/10/2026

Practical No.8

Q. Write a Python program to count frequency of characters in a given file.

Program:

```
filename = input("Enter a file name: ")
f1 = open(filename, 'w')
f1.write('Computer science Engineering\n')
f1.write('Electronics and Communication Engineering\n')
f1.write('Civil Engineering\n')
f1.close()
f1 = open(filename, 'r')
data = f1.read()
a = list(set(data))
a.sort()
print(a)
f1.close()
for i in a:
    print("{} as occurred {} times".format(i, data.count(i)))
```

Output:

```
Enter a file name: MCA
['\n', ' ', 'C', 'E', 'a', 'c', 'd', 'e', 'g', 'i', 'l', 'm', 'n', 'o', 'p',
'r', 's', 't', 'u', 'v']

 as occurred 3 times
  as occurred 6 times
C as occurred 3 times
E as occurred 4 times
a as occurred 2 times
c as occurred 5 times
d as occurred 1 times
e as occurred 10 times
g as occurred 6 times
i as occurred 12 times
l as occurred 2 times
m as occurred 3 times
n as occurred 14 times
o as occurred 4 times
p as occurred 1 times
r as occurred 5 times
s as occurred 2 times
t as occurred 3 times
u as occurred 2 times
v as occurred 1 times
```